$Var(ax+by) = a^2 var(x) + b^2 var(y) + 2ab\,cov(xy)$

HW6

1) a) X_1: quantitative observation data point
Y: Dependent variable
$(1|X_2)$ & $(1|X_3)$: because X_2 & X_3 are two different methods of sampling the group IDs (example schoolID vs. neighborhood ID), these represent random effects, where the method of sampling #1 (corresponding to X_2) might affect the output differently than sampling method 2, but we can still treat them as random LVL 2 factors & interpret their coefficients

b) For unconditional Y, or EBLUP means will be 0 since these random effects are typically sampled from $\sim N(0,\sigma^2)$, so $E(\hat{Y}) = \beta_0 + \beta_1 X_1$
Because we assume that X_2 & X_3 are sampled from independent normal models, we can just add the variances and take the sqrt to get sd
$SD(\sqrt{\sigma^2_{X_2} + \sigma^2_{X_3} + \sigma^2})$ & where σ^2 = residual variance
For conditional, the $E(Y|Y=y)$ adds in the EBLUPs, so we get
$E(Y) = \beta_0 + \beta_1 X_1 + EBLUP_{X_2} + EBLUP_{X_3}$
For the conditional, we now in essence fix X_2 & X_3 since our conditional $\hat{Y}$ assumes that we use or given X_2 & X_3, so this variance disappears. We're left w/
$SD(Y|conditional) = \sqrt{\sigma^2}$ & residual variance

c) X_1: quantitative $(X_1|X_2)$ we now have the random effects of crossing X_1 w/ X_2, one of the sampling methods. Instead of this term now being a random intercept, it is now a random slope. The coefficient tells us that "the same change in X_1 could affect the output Y differently based on X_2": Has unique intercept & slope

d) $E(Y) = \beta_0 + \beta_1 X_1$, $SD(Y\ unconditional) = \sqrt{var(Y)}$, where
$Var(Y) = \sigma^2 + 1^2\sigma^2_{\substack{intercept\ or\\ interaction}} + X_1^2\sigma^2_{slope} + 2X_1 cov(b_0, b_1)$ where b_0 & b_1 are the random intercept & slope vars

Conditional:
$E(Y\ conditional) = \beta_0 + \beta_1 X_1 + EBLUP_{b_0} + EBLUP_{b_1} X_1$, $sd(Y)_{conditional} = \sqrt{\sigma^2}$, again we fix our

2) a) To test if treatment random effects are significant, we compare M1 to M2 since M2 contains the treatment random factor & M1 is the upper's nested model not containing this. Since we're discussing factors that are random, we use the REML estimates.

H_0: M_1 sufficient ← more broad model

H_a: M_2 is better

$D = -2(\log REML_{M_1} - \log REML_{M_2}) = 26.0918$ We typically use χ^2_{df} for this test, however since we're testing if a covariance parameter lies on the boundary space of our D matrix (that is, 0), we need to use a test statistic = $.5\chi^2_1 + .5\chi^2_2$. At $\alpha = .05$, our critical values are 3.84 & 5.99, which 26.0918 is significantly above both. We have sufficient statistical evidence to reject the null hypothesis in favor of the alternative that M_2 is a better model.

b) H_0: M_2 (homogeneous) is sufficient

H_a: M_3 (separate) is better

Use REML because our σ^2_{treat} & $\sigma^2_{control}$ affect our random effect D matrix. We are testing at D matrix shapes here.

$D = -2(\log REML_{M_2} - \log REML_{M_3}) = .152 \sim \chi^2_1$ (one df difference), our critical value is @3.84, so there is not enough evidence to prove we should use separate σ^2 for treat & control.

c) H_0: M_0 suff.

H_a: M_1 better

Use ML because dealing w/ fixed effects

$D = -2(\log ML_{M_0} - \log ML_{M_1}) = 25.5834 \sim \chi^2_1$ critical value @3.84 for $\alpha = .05$, pvalue $= 4.2 \times 10^{-7}$. Since our p-value is much below the cutoff of .05, we reject the null to claim that the interaction term is highly significant

d) M_2 has the 2nd model because its ML ~~to~~ REML difference is negligible to that of M_3, and its simplicity makes it easily explainable.

e) Overall this Q-Q plot looks decent. Our points are close to fitting a straight line (meaning they are sampled from a theoretically normal distribution) and towards the tails we do not deviate from this much. We could argue that things get a little squirly around -20 & 0 on the residuals scale, but overall this looks like a decent fit

f) Our points seem to follow a star pattern ( ), which this could be or lack of data (we only have 6 rats), but I don't think this looks randomness. We have a wide spread of residuals toward the middle of our predicted values range, and a more narrow spread at the ends.

g) $D = \begin{bmatrix} \sigma^2_{intercept} & \sigma_{int,trat} \\ \sigma_{int,trat} & \sigma^2_{trat} \end{bmatrix} = \begin{bmatrix} 35.83^2 & .8 \times 35 \times 79 \\ .8 \times 35 \times 79 & 79.82^2 \end{bmatrix} = \begin{bmatrix} 1283.79 & 2291.27 \\ 2291.27 & 6371.23 \end{bmatrix}$

$R_{ij} = I_n \times 23.21432^2$

Conditional:

2) $\hat{y}_{ij} = X_{ij}\hat{\beta} + Z_{ij}\hat{b}_i \rightarrow \hat{y}_{ij} = 212.294 + 360.026\, t_{ij} +$

$\begin{cases} 0 \text{ if rat in control} \\ 1 \text{ if rat in treatment} \end{cases}$

Likely, this observation was R4 in the treatment group

$(1-t_{ij}) \times \begin{cases} -40.93 \text{ if R1} \\ -32.20 \text{ if R2} \\ 29.78 \text{ if R3} \\ 12.24 \text{ if R4} \\ 31.11 \text{ if R5} \end{cases} + t_{ij} \begin{cases} -94.35 \text{ if R1} \\ -44.3 \text{ if R2} \\ 113.62 \text{ if R3} \\ -2.7 \text{ if R4} \\ 27.73 \text{ if R5} \end{cases}$

Unconditional:

$\hat{y}_{ij} = 212.294 + 360.026\, t_{ij} \rightarrow \begin{cases} 0 \text{ if control} \\ 1 \text{ if trat} \end{cases}$

All indicator dummy variables

3) $\hat{y}_{ij} = 212.294 + 216.212(region1_{ij}) + 25.45(region2_{ij}) + 360.026(trat_{ij})$
$- 261.82(reg1{:}trat_{ij}) - 162.5(reg2{:}trat_{ij}) + R_{ij} \rightarrow R_{ij} \sim N(0, 23.21^2)$ } fixed

random $\{ + b_{0i} + b_{1i} \rightarrow \begin{bmatrix} b_{0i} \\ b_{1i} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, D\right)$

as mentioned in pt. 1

3) As our y-pred increases our mean increases, but so does our sd. Because our sd increases, we have reason to suggest that our data follows a heteroskedastic trend, as we increase in y our sd gets bigger, meaning our model may not be the best fit, as it cannot capture the increase in variance associated with increase in y. If this is a linear model (since we've been working w/ them), since the residual means for each group are increasing, our residuals do not appear to follow a Normal distribution, which could indicate that we are missing a crucial factor in our model. The jump in y-pred from 71->76 is a little alarming, meaning we have no data in the middle of those two groups, but our model choice likely won't fix that.

4) M0's residual vs. fitted plot has a clear upward trend where all our residuals are >0. It is clear that countries denoted by red & pink have similar values, as does blue & purple, and burnt orange, green, & teal. I would explore if there are any other factors we could add to M1 to try to make the residuals more consistent instead of increasing as predicted values increase. The residuals seem to fan out a little bit, also possibly suggesting a slight bit of heteroskedasticity in our data. We could also scale our data by some log factor to try to decrease the spread of the residuals in this plot.